

0801 2

一、 1.D 2.C 3.A 4.B 5.A

二、 1. 1.33(或者填 $\frac{1359}{1024}$) 2. 25 3. 0.25 4. $(X+Y) \sim B(7, p)$ 5. 2/5 6. 0.35 7. 相关

三、

设 A = “取出的产品是正品”；

$B_{甲}$ = “取出的产品是甲厂生产的”

$B_{乙}$ = “取出的产品是乙厂生产的”

$B_{丙}$ = “取出的产品是丙厂生产的”

则 $P(A) = P(AB_{甲}) + P(AB_{乙}) + P(AB_{丙})$

$$= 0.5 \times 0.9 + 0.3 \times 0.8 + 0.2 \times 0.7 = 0.83$$

$$P(B_{甲} | A) = \frac{P(AB_{甲})}{P(A)} = \frac{P(B_{甲}) \cdot P(A | B_{甲})}{P(A)} = \frac{0.5 \times 0.9}{0.83} = 0.54$$

四、

$$X \sim \begin{pmatrix} -1 & 1 & 3 \\ 0.3 & 0.5 & 0.2 \end{pmatrix}$$

$$EX = (-1) \times 0.3 + 1 \times 0.5 + 3 \times 0.2 = 0.8$$

五、

$$(1) \text{由题意} f(x) = \begin{cases} \frac{1}{2}e^x & x \leq 0 \\ \frac{1}{2}e^{-x} & x > 0 \end{cases}$$

$$\text{故} F(x) = \begin{cases} \frac{1}{2}e^x & x \leq 0 \\ 1 - \frac{1}{2}e^{-x} & x > 0 \end{cases}$$

$$(2) P(-5 < X < 10) = (1 - \frac{1}{2}e^{-10}) - (\frac{1}{2}e^{-5})$$

$$(3) EX = \int_{-\infty}^0 x \cdot \frac{1}{2}e^x dx + \int_0^{+\infty} x \cdot \frac{1}{2}e^{-x} dx = \frac{1}{2}[(x-1)e^x]_{-\infty}^0 + \frac{1}{2}[(-x-1)e^{-x}]_0^{+\infty} = 0$$

$$EX^2 = \int_{-\infty}^0 x^2 \cdot \frac{1}{2}e^x dx + \int_0^{+\infty} x^2 \cdot \frac{1}{2}e^{-x} dx$$

$$= \frac{1}{2}[x^2 e^x - 2xe^x + 2e^x]_{-\infty}^0 + \frac{1}{2}[-x^2 e^{-x} - 2xe^{-x} - 2e^{-x}]_0^{+\infty} = 2$$

$$DX = EX^2 - (EX)^2 = 2$$

六、

设 X_i 为第 i 台机床生产的次品率

$$X_i \sim U(0.005, 0.035)$$

$$EX_i = \frac{0.005 + 0.035}{2} = 0.02, DX_i = \frac{1}{12}(0.035 - 0.005)^2 = 0.000075$$

(注: 对于均匀分布 $X_i \sim U(a, b)$, 有 $EX_i = \frac{a+b}{2}$, $DX_i = \frac{1}{12}(b-a)^2$)

$$\text{设总次品率 } Y = \sum_{i=1}^{2000} X_i$$

若要满足这批产品的平均次品率小于 0.025, 则 $Y < 0.025 \times 2000 = 50$

$$P\{Y < 50\} = P\left\{\frac{Y - 2000 \times 0.02}{\sqrt{2000 \times 0.000075}} < \frac{50 - 2000 \times 0.02}{\sqrt{2000 \times 0.000075}}\right\} = \Phi(25.8)$$

七、

$$(1) S_{\text{椭圆}} = \pi ab$$

$$\text{故 } (x, y) \text{ 的联合密度函数 } f(x, y) = \begin{cases} \frac{1}{\pi ab} & -a \leq x \leq a, -b \leq y \leq b \\ 0 & \text{其它} \end{cases}$$

$$X \text{ 的边缘密度函数 } f_X(x) = \begin{cases} \frac{2}{\pi a} & -a \leq x \leq a \\ 0 & \text{其它} \end{cases}$$

$$Y \text{ 的边缘密度函数 } f_Y(y) = \begin{cases} \frac{2}{\pi b} & -b \leq y \leq b \\ 0 & \text{其它} \end{cases}$$

$$(2) EX = \int_{-a}^a x \cdot \frac{2}{\pi a} dx = 0, EY = \int_{-b}^b y \cdot \frac{2}{\pi b} dy = 0,$$

$$EX^2 = \int_{-a}^a x^2 \cdot \frac{2}{\pi a} dx = \frac{4a^2}{3\pi}, EY^2 = \int_{-b}^b y^2 \cdot \frac{2}{\pi b} dy = \frac{4b^2}{3\pi}$$

$$DX = EX^2 - (EX)^2 = \frac{4a^2}{3\pi} = 25, DY = EY^2 - (EY)^2 = \frac{4b^2}{3\pi} = 4$$

$$\text{解得 } a = \frac{5\sqrt{3\pi}}{2}, b = \sqrt{3\pi}$$

(3) $-a \leq x \leq a, -b \leq y \leq b$ 时,

$$f_X(x) \cdot f_Y(y) = \frac{2}{\pi a} \cdot \frac{2}{\pi b} \neq \frac{1}{\pi ab}, \text{ 故 } X \text{ 与 } Y \text{ 不独立}$$

八、

$$Z \text{ 的分布函数 } F(z) = P\{Z \leq z\} = 1 - P(Z > z) = 1 - P\{\min(X, Y) > z\}$$

$$= 1 - P(X > z, Y > z) = 1 - P(X > z) \cdot P(Y > z)$$

$$\text{当 } z \leq 0 \text{ 时, } P(X > z) = P(Y > z) = 1$$

$$\text{故 } F(z) = 1 - 1 = 0$$

$$\text{当 } 0 < z \leq 1 \text{ 时, } P(X > z) = \int_z^1 1 dx = 1 - z$$

$$P(Y > z) = \int_z^1 5e^{-5x} dx = e^{-5z} - e^{-5}$$

$$\text{故 } F(z) = 1 - (1 - z) \cdot (e^{-5z} - e^{-5})$$

$$\text{当 } z > 1 \text{ 时, } P(X > z) = 0$$

$$\text{故 } F(z) = 1 - 0 = 1$$

$$\text{所以 } F(z) = \begin{cases} 0 & z \leq 0 \\ 1 - (1 - z) \cdot (e^{-5z} - e^{-5}) & 0 < z \leq 1 \\ 1 & z > 1 \end{cases}$$

$$f(z) = \begin{cases} 0 & z \leq 0 \\ 6e^{-5z} - 5ze^{-5z} - e^{-5} & 0 < z \leq 1 \\ 0 & z > 1 \end{cases}$$

一、解：设 $A_i = \{\text{第 } i \text{ 枚弹道导弹击沉航空母舰}\}$, $B_i = \{\text{第 } i \text{ 枚弹道导弹击伤航空母舰}\}$

$C_i = \{\text{第 } i \text{ 枚弹道导弹没有击中航空母舰}\}$, $i=1, 2, 3, 4$

$D = \{\text{发射 4 枚弹道导弹能击沉航空母舰}\}$

$$P(A_i) = \frac{1}{3}, \quad P(B_i) = \frac{1}{2}, \quad P(C_i) = \frac{1}{6}, \quad i=1, 2, 3, 4$$

$$\bar{D} = C_1 C_2 C_3 C_4 \cup B_1 C_2 C_3 C_4 \cup C_1 B_2 C_3 C_4 \cup C_1 C_2 B_3 C_4 \cup C_1 C_2 C_3 B_4$$

$$\begin{aligned} P(\bar{D}) &= P(C_1 C_2 C_3 C_4) + P(B_1 C_2 C_3 C_4) + P(C_1 B_2 C_3 C_4) + P(C_1 C_2 B_3 C_4) + P(C_1 C_2 C_3 B_4) \\ &= \left(\frac{1}{6}\right)^4 + 4 \times \left(\frac{1}{6}\right)^3 \times \frac{1}{2} = \frac{13}{6^4} \end{aligned}$$

$$P(D) = 1 - P(\bar{D}) = 1 - \frac{13}{6^4} = 0.99$$

二、解：(1) $A = \{\text{同花顺 (5 张同一花色连续数字构成)}\}$

$$P(A) = \frac{4 \times (13-4)}{C_{52}^5} = \frac{36}{C_{52}^5} \quad (\text{只要说明顺子的构成, 分子 40 也算对})$$

(2) $A = \{\text{3 张带一对 (3 张数字相同、2 张数字相同构成)}\}$

$$P(A) = \frac{C_{13}^1 C_4^3 C_{12}^1 C_4^2}{C_{52}^5}$$

(3) $A = \{\text{3 张带 2 散牌 (3 张数字相同、2 张数字不同构成)}\}$

$$P(A) = \frac{C_{13}^1 C_4^3 C_{12}^2 C_4^1 C_4^1}{C_{52}^5}$$

三、解：(1) 设 $A = \{\text{被查后认为是非危险人物}\}$, $B = \{\text{过关的人是危险人物}\}$, 则

$$P(A) = P(B)P(A|B) + P(\bar{B})P(A|\bar{B}) = 0.96 \times 0.98 + 0.04 \times 0.05 = 0.9428$$

$$P(B|A) = \frac{P(B)P(A|B)}{P(A)} = 0.998$$

(2) 设需要 n 道卡, 每道检查系统是相互独立的, 则

$C_i = \{\text{第 } i \text{ 关危险人物被误认为非危险人物}\}$, $P\{C_1 \cdots C_n\} = 0.05^n$, 所以

$$1 - 0.05^n \geq 0.999, \quad n \geq \frac{\ln 0.0001}{\ln 0.05}, \quad \text{即 } n = \left\lceil \frac{\ln 0.0001}{\ln 0.05} \right\rceil + 1 = [3.0745] + 1 = 4$$

四、解：当 $a=1$ 时, $Y=1$, 则 $F_Y(y) = \begin{cases} 0 & y \leq 1 \\ 1 & y > 1 \end{cases}$

$$\text{当 } 0 < a < 1 \text{ 时, 当 } y \leq 0 \text{ 时, } F_Y(y) = P(Y < y) = 0, \quad f_Y(y) = \frac{dF_Y(y)}{dy} = 0$$

$$\text{当 } y > 0 \text{ 时, } F_Y(y) = P(a^X < y) = P(X \ln a < \ln y)$$

$$F_Y(y) = P\left(X > \frac{\ln y}{\ln a}\right) = 1 - P\left(X \leq \frac{\ln y}{\ln a}\right) = 1 - \Phi\left(\frac{\ln y}{\ln a}\right)$$

$$f_Y(y) = \frac{dF_Y(y)}{dy} = -\frac{1}{y \ln a} \cdot \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(\frac{\ln y}{\ln a} - \mu)^2}{2\sigma^2}}$$

$$\text{当 } a > 1 \text{ 时, 当 } y \leq 0 \text{ 时, } F_Y(y) = P(Y < y) = 0, \quad f_Y(y) = \frac{dF_Y(y)}{dy} = 0$$

$$\text{当 } y > 0 \text{ 时, } F_Y(y) = P\left(X < \frac{\ln y}{\ln a}\right) = \Phi\left(\frac{\ln y}{\ln a}\right)$$

$$f_Y(y) = \frac{dF_Y(y)}{dy} = \frac{1}{y \ln a} \cdot \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(\frac{\ln y}{\ln a} - \mu)^2}{2\sigma^2}}$$

五、解：(1) $E(X+Y)=$

$$E(X+Y) = -3a - 2 \times 0.14 - b - 1 \times 0.01 + 1 \times 0.03 + 1 \times 0.13 + 2 \times 0.14 + 3 \times 0.15 \\ = -3a - b + 0.6 = 0$$

$$a + 0.14 + b + 0.01 + 0.02 + 0.03 + 0.12 + 0.13 + 0.14 + 0.15 = a + b + 0.74 = 1$$

联立解得： $a = 0.17$, $b = 0.09$

(2) X 的概率分布函数：

	-2	-1	0	1
	0.17	0.23	0.06	0.54

$$(3) E(XY) = 2 \times 0.17 + 1 \times 0.14 - 1 \times 0.12 + 1 \times 0.14 + 2 \times 0.15 = 0.8$$

$$\text{六、解： } P\left\{\left|\frac{m}{n} - p\right| < 0.1\right\} \geq 0.95, \quad \text{因 } \frac{\frac{m}{n} - p}{\sqrt{\frac{p(1-p)}{n}}} \sim N(0,1)$$

$$P\left\{\frac{\left|\frac{m}{n} - p\right|}{\sqrt{\frac{p(1-p)}{n}}} < \frac{0.1}{\sqrt{\frac{p(1-p)}{n}}}\right\} \geq 0.95, \quad \frac{0.1}{\sqrt{\frac{p(1-p)}{n}}} \geq u_{0.975} = 1.96$$

$$n \geq (1.96)^2 p(1-p); \quad \text{因为 } p(1-p) \leq 1/4, \quad \text{取 } n \geq (1.96)^2 / 4 = 96.04 \text{ 即 } n = 97$$

七、解：（1）二维随机变量(X,Y)的联合概率密度：

$$f(x,y) = \begin{cases} 1/ab, & 0 < x < a, 0 < y < b \\ 0, & \text{others} \end{cases}$$

$$\text{边缘概率密度: } f_X(x) = \begin{cases} 1/a, & 0 < x < a \\ 0, & \text{others} \end{cases}, \quad f_Y(y) = \begin{cases} 1/b, & 0 < y < b \\ 0, & \text{others} \end{cases}$$

$$(2) \quad DX = (1/12)a^2 = 12, DY = (1/12)b^2 = 36, \quad a = 12, b = 12\sqrt{3}$$

(3) 随机变量 X 与 Y 相互独立，因为 $f(x,y) = f_X(x)f_Y(y)$

$$\text{八、解: } P(|\xi| > t) = \int_{|x|>t} dF(x) \leq \int_{|x|>t} \frac{|x|^3}{t^3} dF(x) \leq \int_{|x|\geq 0} \frac{|x|^3}{t^3} dF(x) = \frac{E|\xi|^3}{t^3} = \frac{c}{t^3}$$

$$\text{九、解: (1) } \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x,y) dx dy = \int_0^1 \left(\int_0^1 Axy dy \right) dx = \frac{A}{4} = 1, \quad A=4$$

$$(2) \quad P(X < 0.4, Y < 1.3) = \int_0^{0.4} \left(\int_0^1 4xy dy \right) dx = 0.16$$

$$(3) \quad Ee^{tX+sY} = \int_0^1 \left(\int_0^1 e^{tx+sy} 4xy dy \right) dx = \int_0^1 e^{tx} 4x \left(\left(\frac{ye^{sy}}{s} \right) \Big|_0^1 - \frac{1}{s} \int_0^1 e^{sy} dy \right) dx$$

$$= 4 \left(\frac{e^s}{s} - \frac{e^s}{s^2} + \frac{1}{s^2} \right) \left(\frac{e^t}{t} - \frac{e^t}{t^2} + \frac{1}{t^2} \right)$$

$$(4) \quad EX = \int_0^1 \left(\int_0^1 4x^2 y dy \right) dx = \frac{2}{3}, \quad EX^2 = \int_0^1 \left(\int_0^1 4x^3 y dy \right) dx = \frac{1}{2}$$

$$DX = EX^2 - (EX)^2 = \frac{1}{2} - \frac{4}{9} = \frac{1}{9}, \quad E(XY) = \int_0^1 \left(\int_0^1 4x^2 y^2 dy \right) dx = \frac{4}{9}$$

$$\text{Cov}(X,Y) = EXY - EX \cdot EY = \frac{4}{9} - \frac{2}{3} \times \frac{2}{3} = 0$$

十、解：（1）设 ξ 表示该观众答对题数， $\xi = 0, 1, 2, \dots$

则第 $\xi+1$ 次解答答错（即首次出错）。

答对一题的概率为

$$P(\text{答对题}) = P(\text{答对A题} | \text{选择A题})P(\text{选择A题}) + P(\text{答对B题} | \text{选择B题})P(\text{选择B题})$$

$$= 0.4 \times 0.5 + 0.6 \times 0.5 = 0.5$$

答错一题的概率为 0.5

$$\text{所以 } P(\xi = k) = 0.5^k \times 0.5 = 0.5^{k+1}; \quad E\xi = \sum_{k=0}^{\infty} k \cdot 0.5^{k+1} = 1$$

(2) 观众得到奖励金额 η 的期望值：

$$\text{令 } X = \begin{cases} 1, & \text{答对 } A \text{ 题} \\ 2, & \text{答对 } B \text{ 题} \\ 3, & \text{答错题} \end{cases}, \text{ 则 } X \sim \begin{pmatrix} 1 & 2 & 3 \\ 0.2 & 0.3 & 0.5 \end{pmatrix},$$

$$E\eta = E(E(\eta | X)) = 0.2 \times E(1000 + \eta) + 0.3 \times E(500 + \eta) + 0.5 \times 0$$

$$\therefore E\eta = 700$$

或：答对一题得到奖金的期望为： $0.5 \times 0.4 \times 1000 + 0.5 \times 0.6 \times 500 = 350$

进入第 k 题答题环节的概率为： 0.5^{k-1}

因此，总奖金的期望为： $\sum_{k=1}^{\infty} 350 \times 0.5^{k-1} = 700$

一、解：(1) $p = 2 \times \frac{4!}{5!} = \frac{2}{5}$ (2) $p = 2 \times \frac{3!}{5!} = \frac{1}{10}$ (3) $p = \frac{2}{5} + \frac{2}{5} - \frac{1}{10} = \frac{7}{10}$

(4) $p = 1 - \frac{7}{10} = \frac{3}{10}$ (5) $p = 1 \times \frac{4!}{5!} = \frac{1}{5}$

二、解 (1) X 的可能值为 0,1,2,3, 所以 X 的概率分布为

$$P(X=k) = \frac{C_3^k C_3^{3-k}}{C_6^3} \quad (k=0,1,2,3)$$

即

X	0	1	2	3
P	$\frac{1}{20}$	$\frac{9}{20}$	$\frac{9}{20}$	$\frac{1}{20}$

因此

$$EX = 0 \times \frac{1}{20} + 1 \times \frac{9}{20} + 2 \times \frac{9}{20} + 3 \times \frac{1}{20} = \frac{3}{2}$$

(2) 设 $A = \{\text{从乙箱中任取一件产品是次品}\}$, 根据全概率公式有

$$P(A) = \sum_{k=0}^3 P\{X=k\} P\{A|X=k\} = \frac{1}{20} \times 0 + \frac{9}{20} \times \frac{1}{6} + \frac{9}{20} \times \frac{2}{6} + \frac{1}{20} \times \frac{3}{6} = \frac{1}{4}$$

三、解 设 $\xi_i = \begin{cases} 1 & \text{第 } i \text{ 台电视机坏} \\ 0 & \text{第 } i \text{ 台电视机正常} \end{cases}$
 $i = 1, 2, \dots, 9000$

$$P\{\xi_i = 1\} = 0.001 \quad P\{\xi_i = 0\} = 0.999$$

$$E\xi_i = 0.001 \quad D\xi_i = 0.000999$$

$$\sum_{i=1}^{9000} E\xi_i = 9 \quad \sum_{i=1}^{9000} D\xi_i \approx 9$$

保险公司亏, 则电视机坏的台数:

$$> 9000 \times 5 / 2000 = 22.5$$

$$P\left\{\sum_{i=1}^{9000} \xi_i > 22.5\right\} = P\left\{\frac{\sum_{i=1}^{9000} \xi_i - E\left(\sum_{i=1}^{9000} \xi_i\right)}{\sqrt{D\left(\sum_{i=1}^{9000} \xi_i\right)}} > \frac{22.5 - 9}{\sqrt{9}}\right\} = 1 - \Phi(4.5) \approx 0$$

保险公司获利不少于 10000 元, 则电视机坏的台数:

$$< (9000 \times 5 - 10000) / 2000 = 17.5$$

$$P\left\{\sum_{i=1}^{9000}\xi_i < 17.5\right\} = P\left\{\frac{\sum_{i=1}^{9000}\xi_i - E\left(\sum_{i=1}^{9000}\xi_i\right)}{\sqrt{D\left(\sum_{i=1}^{9000}\xi_i\right)}} < \frac{17.5-9}{\sqrt{9}}\right\} = \Phi(2.83)$$

$$= \Phi(2) + \frac{\Phi(3) - \Phi(2)}{3-2}(2.83-2) = 0.9772 + 0.02145 \times 0.83 = 0.995$$

四、解 (1)由概率分布的性质可知, $a+b+c+0.6=1$,即 $a+b+c=0.4$.

由 $EX = -0.2$,可得 $-a+c = -0.1$.

$$\text{再由 } P\{Y \leq 0 | X \leq 0\} = \frac{P\{X \leq 0, Y \leq 0\}}{P\{X \leq 0\}} = \frac{a+b+0.1}{a+b+0.5} = 0.5, \text{解得 } a+b=0.3.$$

解以上关于 a 、 b 、 c 的三个方程可得, $a=0.2, b=0.1, c=0.1$.

(2) Z 的所有可能取值为-2,-1,0,1,2.则

$$P\{Z = -2\} = P\{X = -1, Y = -1\} = 0.2$$

$$P\{Z = -1\} = P\{X = -1, Y = 0\} + P\{X = 0, Y = -1\} = 0.1$$

$$P\{Z = 0\} = P\{X = -1, Y = 1\} + P\{X = 1, Y = -1\} + P\{X = 0, Y = 0\} = 0.3$$

$$P\{Z = 1\} = P\{X = 1, Y = 0\} + P\{X = 0, Y = 1\} = 0.3$$

$$P\{Z = 2\} = P\{X = 1, Y = 1\} = 0.1$$

所以 Z 的概率分布为

Z	-2	-1	0	1	2
P	0.2	0.1	0.3	0.3	0.1

$$(3) P\{X = Z\} = P\{Y = 0\} = 0 + b + 0.1 = 0.1 + 0.1 = 0.2.$$

五、解 (1) Y 的分布函数为 $F_Y(y) = P\{Y \leq y\} = P\{X^2 \leq y\}$

当 $y \leq 0$ 时, $F_Y(y) = 0, f_Y(y) = 0$.

当 $0 < y < 1$ 时,

$$F_Y(y) = P\{-\sqrt{y} \leq X \leq \sqrt{y}\} = P\{-\sqrt{y} \leq X < 0\} + P\{0 \leq X \leq \sqrt{y}\} = \frac{3}{4}\sqrt{y}$$

$$f_Y(y) = \frac{3}{8\sqrt{y}}$$

$$\text{当 } 1 \leq y < 4 \text{ 时, } F_Y(y) = P\{-1 \leq X < 0\} + P\{0 \leq X \leq \sqrt{y}\} = \frac{1}{2} + \frac{1}{4}\sqrt{y}$$

$$f_Y(y) = \frac{1}{8\sqrt{y}}$$

$$\text{当 } y \geq 4 \text{ 时, } F_Y(y) = 1, f_Y(y) = 0.$$

$$\text{所以 } Y \text{ 的概率密度为 } f_Y(y) = \begin{cases} \frac{3}{8\sqrt{y}} & \text{当 } 0 < y < 1 \\ \frac{1}{8\sqrt{y}} & \text{当 } 1 \leq y < 4 \\ 0 & \text{其他} \end{cases}$$

$$(2) \quad EX = \int_{-\infty}^{+\infty} xf_X(x)dx = \int_{-1}^0 \frac{1}{2}xdx + \int_0^2 \frac{1}{4}xdx = \frac{1}{4}$$

$$EY = EX^2 = \int_{-\infty}^{+\infty} x^2 f_X(x)dx = \int_{-1}^0 \frac{1}{4}x^2dx = \frac{5}{6}$$

$$EXY = EX^3 = \int_{-\infty}^{+\infty} x^3 f_X(x)dx = \int_{-1}^0 \frac{1}{2}x^3dx + \int_0^2 \frac{1}{4}x^3dx = \frac{7}{8}$$

$$\text{故 } \operatorname{cov}(X, Y) = EXY - EX \cdot EY = \frac{2}{3}$$

$$(3) \quad F\left(\frac{1}{2}, 4\right) = P\left\{X \leq -\frac{1}{2}, Y \leq 4\right\} = P\left\{X \leq -\frac{1}{2}, X^2 \leq 4\right\}$$

$$= P\left\{X \leq -\frac{1}{2}, 2 \leq X \leq 2\right\} = P\left\{-2 \leq X \leq -\frac{1}{2}\right\} = P\left\{-1 \leq X \leq -\frac{1}{2}\right\} = \frac{1}{4}$$

六、(2学分) 解 设 $F(y)$ 是 Y 的分布函数, 则由全概率公式可知, $U = X + Y$ 的分布函数为 $G(u) = P\{X + Y \leq u\}$

$$= 0.3P\{X + Y \leq u | X = 1\} + 0.7P\{X + Y \leq u | X = 2\}$$

$$= 0.3P\{Y \leq u - 1 | X = 1\} + 0.7P\{Y \leq u - 2 | X = 2\}$$

由于 X 与 Y 独立, 得

$$G(u) = 0.3P\{Y \leq u - 1\} + 0.7P\{Y \leq u - 2\} = 0.3F(u - 1) + 0.7F(u - 2)$$

因此, U 的概率密度为

$$g(u) = G'(u) = 0.3F'(u - 1) + 0.7F'(u - 2) = 0.3f(u - 1) + 0.7f(u - 2)$$

七、(2学分) 解 设 $A = \{\text{抽到一名男性}\}$; $B = \{\text{抽到一名女性}\}$; $C = \{\text{抽到一名色盲患者}\}$, 由全概率公式得

$$P(C) = P(C | A)P(A) + P(C | B)P(B) = 5\% \times \frac{1}{2} + 0.25\% \times \frac{1}{2} = 2.625\%$$

$$P(AC) = P(A)P(C|A) = \frac{1}{2} \times 5\% = 2.5\%$$

由贝叶斯公式得 $P(A|C) = \frac{P(AC)}{P(C)} = \frac{20}{21}$

八、(2 学分) 解 (1) $Y_1 + Y_n = X_1 - \bar{X} + X_n - \bar{X} = X_1 - \frac{2}{n} \sum_{i=2}^{n-1} X_i + X_n$

上式是相互独立的正态随机变量的线性组合, 所以 $Y_1 + Y_n$ 服从正态分布.

由于 $E(Y_1 + Y_n) = 0$, 故 $P\{Y_1 + Y_n \leq 0\} = \frac{1}{2}$

(2) 将 b 只白球, c 只黑球看成 $b+c$ 只非红球, 则与两种颜色球问题相同. 用归纳法可得第 k 次取到红球的概率为: $p_k = \frac{a}{a+b+c}$

六、(3、4 学分)

解 (1) $DY_i = D(X_i - \bar{X}) = D\left[\left(1 - \frac{1}{n}\right)X_i - \frac{1}{n} \sum_{k \neq i} X_k\right] = \frac{n-1}{n} \sigma^2 \quad (i = 1, 2, \dots, n)$

(2) $\text{cov}(Y_1, Y_n) = E(Y_1 - EY_1)(Y_n - EY_n) = E(X_1 - \bar{X})(X_n - \bar{X})$

$$= EX_1 X_n + E\bar{X}^2 - EX_1 E\bar{X} - EX_n \bar{X}$$

$$= EX_1 EX_n + D\bar{X} - \frac{1}{n} EX_1^2 - \frac{1}{n} \sum_{i=2}^n EX_1 X_i - \frac{1}{n} EX_n^2 - \frac{1}{n} \sum_{i=1}^{n-1} EX_i X_n$$

$$= -\frac{1}{n} \sigma^2$$

(3) $E[c(Y_1 + Y_n)^2] = cD(Y_1 + Y_n) = c[DY_1 + DY_n + 2\text{cov}(Y_1, Y_n)]$

$$= c\left[\frac{n-1}{n} + \frac{n-1}{n} - \frac{2}{n}\right] \sigma^2 = \frac{2(n-2)}{n} c \sigma^2 = \sigma^2$$

故

$$c = \frac{n}{2(n-2)}$$

(4) 令 $D\hat{\theta}_2 = \lambda D\hat{\theta}_1$, 则 $\hat{\theta}_3 = \frac{\lambda}{1+\lambda} \hat{\theta}_1 + \frac{1}{1+\lambda} \hat{\theta}_2$ 是比 $\hat{\theta}_1$ 与 $\hat{\theta}_2$ 更有效的无偏估计量. $D\hat{\theta}_3 = \frac{\lambda}{1+\lambda} D\hat{\theta}_1 < D\hat{\theta}_1, D\hat{\theta}_2$

七、(3、4 学分) 解 (1) Y 的概率密度为

$$f(y) = \frac{1}{\sqrt{2\pi}} e^{-(y-\mu)^2/2} \quad (-\infty < y < +\infty)$$

令 $t = y - \mu$, 则

$$\begin{aligned} b = EX = Ee^Y &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^y e^{-(y-\mu)^2/2} dy = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{t+\mu} e^{-t^2/2} dt \\ &= e^{\mu+1/2} \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-(t-1)^2/2} dt = e^{\mu+1/2} \end{aligned}$$

(2) 当置信度 $1-\alpha=0.95$ 时, $\alpha=0.05, u_{\alpha/2}=u_{0.025}=1.96$. 故由 $\bar{Y} \sim N(\mu, 1/4)$,

可得
$$P\left\{\bar{Y}-1.96 \times \frac{1}{\sqrt{4}}\right\} < \mu < P\left\{\bar{Y}+1.96 \times \frac{1}{\sqrt{4}}\right\} = 0.95$$

其中
$$\bar{Y} = \frac{1}{4}(\ln 0.5 + \ln 0.8 + \ln 1.25 + \ln 2) = \frac{1}{4} \ln 1 = 0$$

于是有
$$P\{-0.98 < \mu < 0.98\} = 0.95$$

因此, μ 的置信度为 0.95 的置信区间为 $(-0.98, 0.98)$.

(3) 由 e^x 的严格递增性, 可知

$$0.95 = P\left\{-0.48 < \mu + \frac{1}{2} < 1.48\right\} = P\left\{e^{-0.48} < e^{\mu+\frac{1}{2}} < e^{1.48}\right\}$$

因此, b 的置信度为 0.95 的置信区间为 $(e^{-0.48}, e^{1.48})$.

八、(3、4 学分) 解 检验假设 $H_0: \mu = 235.5, H_1: \mu \neq 235.5$.

由于方差 σ^2 未知, 故采用 t 检验法, 其拒绝域为

$$W = \left\{ |t| = \frac{|\sqrt{n}(\bar{X} - \mu_0)|}{S} \geq t_{\alpha/2}(n-1) \right\}$$

已知 $n=49, \bar{X}=236.5, S=3.5$, 计算得

$$|t| = \frac{|\sqrt{n}(\bar{X} - \mu_0)|}{S} = \left| \frac{\sqrt{49}(236.5 - 235.5)}{3.5} \right| = 2$$

由于 $49-1=48 > 30$, 故可用 $u_{\alpha/2}$ 代替 $t_{\alpha/2}(49-1)$.

当 $\alpha=0.01$ 时, $u_{0.005}=2.58 > 2$, 故接受 H_0 . 即本月该市居民平均伙食费较之上个月无显著变化.

一、解: $P(\overline{ABC}) = P(\overline{A \cup B \cup C}) = 1 - P(A \cup B \cup C)$

$$= 1 - (P(A) + P(B) + P(C) - P(AB) - P(AC) - P(BC) + P(ABC)) = \frac{3}{8}$$

$$(P(AC) = 0, \quad P(ABC) = 0)$$

二、解: (1) 设 A: 第二次 6 个球中的第 5 个是红球 $P(A) = \frac{10}{15} = \frac{2}{3}$

(2) 设 A: 第一次 5 个球中有 2 个黄球 B: 第二次 6 个球中有 4 个红球
原问题转换为求 $P(AB)$

$$\textcircled{1}: \Omega: C_{15}^5 \quad AB: C_5^2 \cdot C_6^2 \cdot C_4^1 \quad P(AB) = \frac{C_5^2 \cdot C_6^2 \cdot C_4^1}{C_{15}^5} = \frac{200}{1001} \approx 0.2$$

$$\textcircled{2}: P(AB) = P(A) \cdot P(B|A) = \frac{C_5^2 \cdot C_{10}^3}{C_{15}^5} \cdot \frac{C_3^2 \cdot C_7^4}{C_{10}^6} = \frac{200}{1001} \approx 0.2$$

(3) 设 A: 第一次 5 个球中有 3 个红球 B: 第二次 6 个球中有 2 个黄球
原问题转换为求 $P(A \cup B)$

$$P(A) = \frac{C_5^2 \cdot C_{10}^3}{C_{15}^5}, \quad P(B) = \frac{C_5^2 \cdot C_{10}^4}{C_{15}^6} \left(= \frac{C_6^2 \cdot C_9^3}{C_{15}^5} \right)$$

$$P(AB) = \frac{C_5^2 \cdot C_6^2 \cdot C_4^1}{C_{15}^5}$$

$$P(A \cup B) = P(A) + P(B) - P(AB) = \frac{620}{1001} \approx 0.62$$

$$\text{三、} F_\eta(y) = P(\eta < y) = P(2^\xi < y) = P(\xi < \ln y / \ln 2) = \frac{1}{2\sqrt{2\pi}} \int_{-\infty}^{\frac{\ln y}{\ln 2}} e^{-\frac{x^2}{8}} dx$$

$$f_\eta(y) = F'_\eta(y) = \frac{1}{2\ln 2\sqrt{2\pi}} \cdot \frac{1}{y} \cdot e^{-\frac{\ln^2 y}{8\ln^2 2}}$$

$$(2) E\eta = \frac{1}{2\sqrt{2\pi}} \int_{-\infty}^{+\infty} 2^x \cdot e^{-\frac{x^2}{8}} dx = \frac{1}{2\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-\frac{1}{8}((x-4\ln 2)^2 - 16\ln^2 2)} dx = e^{2\ln^2 2} = 2^{2\ln 2}$$

$$\begin{aligned} (3) D\eta &= E\eta^2 - (E\eta)^2 = \frac{1}{2\sqrt{2\pi}} \int_{-\infty}^{+\infty} 2^{2x} \cdot e^{-\frac{x^2}{8}} dx - e^{4\ln^2 2} \\ &= \frac{1}{2\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-\frac{1}{8}((x-8\ln 2)^2 - 64\ln^2 2)} dx - e^{4\ln^2 2} = e^{8\ln^2 2} - e^{4\ln^2 2} = 2^{4\ln 2} (2^{4\ln 2} - 1) \end{aligned}$$

四、解: 设 A: 产品为次品 B_i : 产品取自第 i 盒, $i=1, 2, 3$

则: $P(B_1)=1/2, P(B_2)=1/6, P(B_3)=1/3$

$$P(A|B_1)=3/9, P(A|B_2)=2/12, P(A|B_3)=6/24$$

$$(1) \quad P(A) = \sum_{i=1}^3 P(AB_i) = \sum_{i=1}^3 P(B_i) \cdot P(A|B_i) = 5/18$$

$$(2) \quad P(B_k|A) = \frac{P(AB_k)}{P(A)} = \begin{cases} \frac{3}{5} & k=1 \\ \frac{1}{10} & k=2 \\ \frac{3}{10} & k=3 \end{cases}$$

五、解：设 ξ_i ：第 i 周的销量，则： $\xi_i \sim P(9)$ ， $i=1 \sim 50$ 。令： $\mu=E\xi_i=9$ ， $\sigma^2=D\xi_i=9$

$$(1) \quad P(400 \leq \sum_{i=1}^{50} \xi_i \leq 500) = P\left(\frac{400 - 50 \cdot \mu}{\sigma \times \sqrt{50}} \leq \frac{\sum_{i=1}^{50} \xi_i - 50 \cdot \mu}{\sigma \times \sqrt{50}} \leq \frac{500 - 50 \cdot \mu}{\sigma \times \sqrt{50}}\right)$$

$$= 2\phi\left(\frac{\sqrt{50}}{3}\right) - 1 = 2\phi(2.36) - 1 = 0.9818$$

$$(2) \quad \text{设：} m \text{ 为最低利润，求 } m, \text{ s.t. } P\left(10 \cdot \sum_{i=1}^{50} \xi_i \geq m\right) \geq 0.95$$

$$P\left(10 \cdot \sum_{i=1}^{50} \xi_i \geq m\right) = 1 - P\left(\sum_{i=1}^{50} \xi_i \leq \frac{m}{10}\right) = 1 - \phi\left(\frac{\frac{m}{10} - 450}{3\sqrt{50}}\right)$$

$$\phi\left(\frac{450 - \frac{m}{10}}{3\sqrt{50}}\right) = 0.95, \quad m = 4151 \text{ 元}$$

$$(3) \quad \text{设：} M \text{ 为最高利润，求 } M, \text{ s.t. } P\left(10 \cdot \sum_{i=1}^{50} \xi_i \leq M\right) \geq 0.95$$

$$P\left(10 \cdot \sum_{i=1}^{50} \xi_i \leq M\right) = P\left(\frac{\sum_{i=1}^{50} \xi_i - 450}{3 \cdot \sqrt{50}} \leq \frac{\frac{M}{10} - 450}{3 \cdot \sqrt{50}}\right) = \phi\left(\frac{\frac{M}{10} - 450}{3\sqrt{50}}\right)$$

$$\phi\left(\frac{\frac{M}{10} - 450}{3\sqrt{50}}\right) = 0.95, \quad M = 4849 \text{ 元}$$

六、(2 学分) 解：设：事件 A：长度误差不小于 3%， $n=100$ ， $p=P(A)$

η ：试生产的 n 个工件中长度误差不小于 3% 的工件个数

则： $\eta \sim B(n, p)$

$$p = P(A) = P(|\xi - 100| \geq 3) = 1 - P\left(-\frac{3}{\sqrt{2.34}} < \frac{\xi - 100}{\sqrt{2.34}} < \frac{3}{\sqrt{2.34}}\right)$$

$$= 2\left(1 - \Phi\left(\frac{3}{\sqrt{2.34}}\right)\right) = 0.05 \quad \lambda = np = 5$$

$$P(\eta \geq 3) = 1 - e^{-5}(1 + 5 + 25/2) = 0.8753$$

七、(2 学分) 解: ① $\int_{-\infty}^{+\infty} dx \int_{-\infty}^{+\infty} \varphi(x, y) dy = A \int_0^{+\infty} dx \int_0^{+\infty} e^{-(2x+3y)} dy = 1, \quad A=6$

② $P((\xi, \eta) \in D) = \iint_{(x,y) \in D} \varphi(x, y) dx dy$

$$= 6 \int_0^1 dx \int_0^{x+\frac{1}{3}} e^{-(2x+3y)} dy + 6 \int_1^3 dx \int_0^{\frac{6-2x}{3}} e^{-(2x+3y)} dy = 1 - \frac{2}{5}e^{-1} - \frac{23}{5}e^{-6}$$

八、(2 学分) 解: ① $P\left\{\xi > \frac{1}{2}\right\} = \int_{\frac{1}{2}}^1 \frac{1}{2} dx + P\{\xi = 1\} + P\{\xi = 2\} + P\{\xi = 3\}$

$$= \frac{1}{4} + \left(\frac{2}{3} - \frac{1}{2}\right) + \left(\frac{11}{12} - \frac{2}{3}\right) + \left(1 - \frac{11}{12}\right) = \frac{3}{4}$$

② $P\{2 \leq \xi \leq 4\} = P\{\xi = 2\} + P\{\xi = 3\} = \frac{1}{4} + \frac{1}{12} = \frac{1}{3}$

③ $E\xi = \int_0^1 x \cdot \frac{1}{2} dx + \sum_{i=1}^3 i \cdot P\{\xi = i\} = \frac{7}{6}$

六、(3、4 学分) 解: 求 n, s.t. $P\left\{|\mu - \bar{X}| \leq 20\right\} \geq 0.95$

$$P\left\{|\mu - \bar{X}| \leq 20\right\} = P\left\{-\frac{20}{\sigma/\sqrt{n}} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq \frac{20}{\sigma/\sqrt{n}}\right\} = \Phi\left(\frac{\sqrt{n}}{5}\right) - \Phi\left(-\frac{\sqrt{n}}{5}\right)$$

$$\Phi\left(\frac{\sqrt{n}}{5}\right) = 0.975 \quad n = 96.04 \quad \text{至少调查 97 家}$$

七、(3、4 学分) 解: 1、 σ 未知, 检验 $H_0: \mu = 100$ ($n=10, \alpha=0.05$)

$$t = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1) = t(9) \quad \text{观察值} = \frac{98.2 - 100}{2.25/\sqrt{10}} = -2.53$$

$$t_{\frac{\alpha}{2}}(n-1) = t_{0.025}(9) = 2.26 \quad \text{拒绝原假设 } H_0: \mu = 100$$

2、 μ 未知, 检验 $H_0: \sigma^2 = \sigma_0^2 = 4$

$$\chi^2 = \frac{1}{\sigma_0^2} \sum_{i=1}^n (\xi_i - \bar{X})^2 = \frac{n-1}{\sigma_0^2} S^2 \sim \chi^2(n-1) = \chi^2(9)$$

$$\text{观察值} = 9 \cdot 2.25^2 / 4 = 11.39$$

$$\chi_{0.975}^2(9) = 2.7, \quad \chi_{0.025}^2(9) = 19 \quad \text{接受 } H_0: \sigma^2 = \sigma_0^2 = 4$$

结论：工作不正常，装袋量偏低。

八、（3、4 学分）解： 1、 $\mu = E\xi = \int_{-\infty}^{+\infty} x \cdot f(x) dx = \int_{\theta}^{+\infty} x \cdot 2e^{-2(x-\theta)} dx = \theta + \frac{1}{2}$

$$\bar{X} = \hat{\theta} + \frac{1}{2} \quad \hat{\theta} = \bar{X} - \frac{1}{2} \quad \text{其中：} \quad \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

2、 $E\hat{\theta} = E\bar{X} - \frac{1}{2} = \mu - \frac{1}{2} = \theta$ $\hat{\theta}$ 是参数 θ 的无偏估计